

Experiment 10: SNMP Enumeration

Aim

To perform SNMP enumeration on an SNMP-enabled network device in order to extract information such as system details, running services, and network interfaces.

Algorithm / Procedure

1. Identify a lab device/VM with SNMP enabled (commonly using the default 'public' community string).
2. Open a terminal on the Kali Linux machine.
3. Use a scanning tool (e.g., Nmap) to confirm SNMP (UDP port 161) is open on the target.
4. Use snmpwalk to query the device using the community string.
5. Extract system information (hostname, uptime, OS description).
6. Extract information about running processes, installed software, and network interfaces.
7. Document the information disclosed and recommend hardening steps (change default community strings, use SNMPv3, restrict access).

command

```
# 1. Check if SNMP (UDP 161) is open  
on the target  
nmap -sU -p 161 127.0.0.1
```

```
# 2. Enumerate system information  
using numerical OID  
snmpwalk -v1 -c public 127.0.0.1  
1.3.6.1.2.1.1
```

Result

The above procedure was successfully executed in the lab environment and the observed output was recorded/verified. (Attach screenshot of your terminal output here.)

Output:

ISO.3.6.1.2.1.1.1.0 = STRING: "linux
Kali 7.1.5+Kali-amd64 #1 SMP
PREEMPT_DYNAMIC Kali 7.1.5-1
Kali 1 (2026-07-29) x 86-64"

ISO.3.6.1.2.1.1.2.0 = OID: ISO.3.6.1.4.1.
8072.3.2.10

ISO.3.6.1.2.1.1.3.0 = TimeTicks: (1393118)
3:52:11.18

ISO.3.6.1.2.1.1.4.0 = STRING: "Me < me@
example.org >"

ISO.3.6.1.2.1.1.5.0 = STRING: "Kali"

ISO.3.6.1.2.1.1.6.0 = STRING: "Sitting on
the Dock of the Bay"

ISO.3.6.1.2.1.1.7.0 = INTEGER: 72.